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Financial Frictions and the Great Productivity Slowdown

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1 Big picture

- **Question.** Did financial frictions generated by the 2008 global financial crisis contribute to the persistent post-crisis productivity slowdown, and if so through what within-firm channel?
- **Setting.** The paper studies nonfarm, nonfinancial firms in a cross-country panel of advanced economies around the 2008 Lehman shock, using long differences between the 6 years before the crisis and the 6 years after it.
- **Core empirical object.** The main treatment is *pre-crisis rollover risk*: the amount of debt scheduled to mature in 2008 relative to pre-crisis sales. The main source of quasi-experimental variation is the sharp, unexpected tightening of credit after Lehman Brothers collapsed.
- **Main finding.** Firms with weaker pre-crisis balance sheets suffered a much larger and highly persistent decline in post-crisis within-firm TFP growth. The effect is stronger when the tightening of credit conditions is more severe, either at the country level or for the firm's own creditor banks.
- **Magnitude.** In the preferred baseline specification, a 10-percentage-point larger share of debt maturing in 2008 is associated with about a 0.94-percentage-point lower annual TFP growth rate after the crisis. A back-of-the-envelope calculation suggests that financial frictions may explain about one-third of the within-firm productivity slowdown in the sample.
- **Mechanism.** Financially fragile firms cut intangible investment and reduce patenting more after the crisis. The paper interprets this as evidence that tighter credit constrained innovation and other hard-to-collateralize investments, creating *TFP hysteresis*.
- **Why we care.** Much of the literature on finance and productivity focuses on misallocation across firms. This paper instead shows that financial crises can also depress productivity *within* firms for a long time, helping explain why banking crises have permanent output costs.
- **Bottom line.** The paper's message is not simply that weak firms shrink more in crises. It is that the interaction of weak balance sheets and a credit supply shock damages firms' innovation and pushes down their productivity path for years.

2 Data and measurement (key objects)

2.1 Core firm panel

- **Source.** The main source is ORBIS, assembled by merging different vintages from Bureau van Dijk.
- **Coverage.** The raw data run from 1998 to 2013. The main analysis keeps firms that exist continuously during the 6 years before and 6 years after the crisis, so the baseline long-difference sample uses 2002–2007 versus 2008–2013.
- **Countries.** The paper focuses on advanced economies with both firm balance-sheet data and bank-credit-risk data: Belgium, Germany, Spain, France, Italy, Japan, Korea, the Netherlands, Portugal, Sweden, and the United Kingdom.
- **Industries.** The sample includes the nonfarm, nonfinancial business sector, covering both manufacturing and services.

- **Cleaning.** Very small firms with fewer than three employees are excluded because of data-reliability concerns.

2.2 Financial vulnerability and credit-tightening measures

- **Preferred vulnerability measure.** The main variable is *Debt maturing 2008*: debt due in 2008 divided by average pre-crisis sales. Empirically this is built from current liabilities at the end of 2007, so it captures rollover risk when credit markets froze.
- **Why this variable is attractive.** Because Lehman was unexpected, the paper argues that firms were unlikely to have chosen their 2007 maturity profile in anticipation of the September 2008 credit shock.
- **Country-level credit shock.** ΔCDS_c is the change in the average CDS spread of domestic banks between the 7 days before and after Lehman’s bankruptcy.
- **Firm-level credit shock.** ΔCDS_i is the average change in the CDS spreads of the firm’s main creditor banks, using the BANKER linkage variable from AMADEUS.
- **Interpretation.** Larger increases in bank CDS spreads proxy for sharper tightening in credit supply.

2.3 Productivity measurement

- **Main outcome.** The main dependent variable is the change in average within-firm TFP growth between the post-crisis and pre-crisis periods.
- **Production-function estimation.** The paper estimates sector-specific production functions using Wooldridge’s (2009) one-step GMM control-function approach.
- **Variables entering production.** Real value added, physical capital, labor, and intermediate inputs are used, with deflators from OECD STAN rebased using country-industry PPPs.
- **Important caveat.** This is revenue-based TFP, so firm-specific prices within country-industry cells can matter. The authors argue that if anything this likely biases results downward because financially constrained firms raised prices in the crisis.

2.4 Innovation proxies

- **Intangible investment.** The first mechanism variable is the change in intangible investment relative to value added.
- **Intangible share.** The second is the change in the share of intangible assets in total fixed assets.
- **Patents.** The third is the change in the average number of annual patent applications, using a firm-level match between ORBIS and the European Patent Office’s Patstat database.

2.5 Descriptive statistics

- **Baseline sample size.** The main baseline regressions use 134,838 firms.

- **Average slowdown.** Mean TFP growth falls from 2.14% pre-crisis to -6.73% post-crisis, a decline of about 10.72 percentage points.
- **Distribution of rollover risk.** The median firm has debt maturing in 2008 equal to roughly 25% of pre-crisis sales, with substantial cross-firm dispersion.
- **Cross-country pattern.** Peripheral euro-area countries have both higher average rollover risk and stronger tightening in bank CDS spreads, foreshadowing stronger effects there.

3 Models and empirical specifications (all equations + interpretation)

3.1 There is no structural model; the paper is reduced-form

The paper is mainly an empirical long-difference difference-in-differences design. The theory is verbal: firms with more maturing debt are more exposed when credit freezes, so they cut hard-to-collateralize investments and suffer persistent productivity losses.

3.2 Baseline long-difference specification

$$\Delta TFP_{isc}^{growth} = \beta_1 Vulnerabilities_i^{pre} + \alpha_{sc} + \gamma' X_i + \varepsilon_{isc}. \quad (1)$$

- **Dependent variable.** $\Delta TFP_{isc}^{growth}$ is the difference between the firm's average annual TFP growth in 2008–2013 and its average annual TFP growth in 2002–2007.
- **Regressor of interest.** $Vulnerabilities_i^{pre}$ is mainly debt maturing in 2008.
- **Fixed effects.** α_{sc} are country-sector fixed effects, so identification comes from comparing more and less vulnerable firms within narrow country-industry cells.
- **Controls.** X_i includes firm age, size, and earnings before the crisis.
- **Expected sign.** If rollover risk matters, then $\beta_1 < 0$.
- **Interpretation.** This is the paper's core within-firm design: firms with more debt coming due right when credit froze should see larger subsequent productivity slowdowns.

3.3 Country-level heterogeneity in credit tightening

$$\Delta TFP_{isc}^{growth} = \beta_1 Vulnerabilities_i^{pre} + \beta_2 (Vulnerabilities_i^{pre} \times \Delta CDS_c) + \alpha_{sc} + \gamma' X_i + \varepsilon_{isc}. \quad (2)$$

- **New regressor.** ΔCDS_c is the change in domestic banks' average CDS spread around Lehman.
- **Prediction.** If the baseline effect is genuinely driven by credit supply, then $\beta_2 < 0$: vulnerable firms should do especially badly where bank funding stress rises more.
- **Interpretation.** This equation turns the basic DD into a sharper interaction design that uses cross-country variation in the severity of the Lehman shock.

3.4 Firm-bank matched exposure to creditor stress

$$\Delta TFP_{isc}^{growth} = \beta_1 Vulnerabilities_i^{pre} + \beta_2 (Vulnerabilities_i^{pre} \times \Delta CDS_i) + \beta_3 \Delta CDS_i + \alpha_{sc} + \gamma' X_i + \varepsilon_{isc}. \quad (3)$$

- **New regressor.** ΔCDS_i is the average Lehman-window CDS change for the firm's main creditor bank(s).
- **Expected sign.** Again, $\beta_2 < 0$.
- **Purpose.** This specification pushes identification closer to within-country firm-level variation in credit-supply tightening.
- **Cost.** Sample size falls sharply because matched-bank data and bank CDS quotes are not available for many firms.

3.5 Construction of firm-level TFP

$$TFP_{icst} = va_{icst} - \hat{\beta}_{k,s} k_{icst} - \hat{\beta}_{\ell,s} \ell_{icst-1}. \quad (4)$$

- **Notation.** va is real value added, k is real physical capital, and ℓ is labor.
- **Implementation.** Sector-specific elasticities are estimated using Wooldridge GMM, with the production-function control-function machinery addressing simultaneity between input choice and productivity.
- **Interpretation.** TFP is the residual component of value added after removing the fitted contributions of observable inputs.

3.6 Mechanism equations: intangible investment and patenting

The paper reuses the same long-difference logic with different dependent variables:

$$\Delta IntInvestment_{isc} = \beta_1 Vulnerabilities_i^{pre} + \alpha_{sc} + \gamma' X_i + \varepsilon_{isc}, \quad (5)$$

$$\Delta ShareIntangible_{isc} = \beta_1 Vulnerabilities_i^{pre} + \alpha_{sc} + \gamma' X_i + \varepsilon_{isc}, \quad (6)$$

$$\Delta PatentApplications_{isc} = \beta_1 Vulnerabilities_i^{pre} + \alpha_{sc} + \gamma' X_i + \varepsilon_{isc}. \quad (7)$$

- **Economic idea.** If financially fragile firms cannot roll over debt, they should cut intangible and innovative activity more than resilient firms.
- **Prediction.** The coefficients should be negative if innovation is one channel linking finance to post-crisis productivity losses.
- **Why this matters.** These regressions move the paper beyond documenting the slowdown and toward explaining its persistence.

4 Identification and instruments (what makes the estimates credible)

4.1 Baseline identification idea

- The shock is the sudden, largely unforeseen tightening of credit conditions after Lehman's bankruptcy on September 15, 2008.
- The main cross-sectional exposure is pre-crisis debt maturing in 2008. Since the crisis timing was not anticipated, the debt profile at the end of 2007 is argued to be plausibly predetermined relative to the shock.
- Country-sector fixed effects mean the comparison is not across broad sectors or countries, but within narrowly defined country-industry cells. This absorbs country-industry demand shocks, trade shocks, reform effects, and differences in financial dependence across sectors.
- The authors also show that more and less vulnerable firms do not differ systematically in pre-crisis productivity trends, which weakens the concern that fragile firms were simply structurally worse firms.

4.2 Why the CDS interactions help

- The country-level CDS measure uses a very narrow event window around Lehman, making it more plausible that the rise in bank risk reflects a credit-supply shock rather than deteriorating real activity.
- If the paper were just picking up weak-demand sectors or bad firms, the effect of debt maturing in 2008 should not systematically become stronger where bank CDS spreads jump more.
- The matched-bank design is even sharper: within a country, firms whose creditor banks were hit harder should be more constrained, and that is exactly where the paper finds a bigger effect.

4.3 Placebo and robustness logic

- **Placebo recession.** Replacing 2008 with 2000 yields no significant effect of debt maturing in 2000 on subsequent TFP or intangible investment. This is important because the 2000 recession was not a banking crisis.
- **Short-term debt concern.** The paper shows that the results survive when excluding trade credit, restricting to firms whose debt had maturity at least two years, or excluding firms with only short-term debt.
- **Alternative productivity measures.** The main result survives when TFP is measured by OLS, by a Cobb-Douglas Solow residual, or by labor productivity.
- **Other vulnerabilities.** Adding cash holdings, leverage, or interest-coverage ratios does not undo the role of debt maturing in 2008.

4.4 What is still not fully nailed down

- The paper cannot directly separate long-term from short-term maturing debt with the ideal precision one would want, because ORBIS does not provide the full maturity decomposition.

- Revenue-based TFP remains an imperfect productivity measure.
- The innovation mechanism is strongly suggestive but not exhaustive: the paper openly notes that working-capital disruptions and other channels may also matter.

5 Tables and figures

5.1 Table 1 and Figure 1: Descriptives and the raw dynamic pattern

Table 1
Summary statistics

	Mean	Median	P25	P75	SD
ΔTFP_{growth}	-10.72	-8.78	-30.54	9.00	33.83
TFP_{pre}^{growth}	2.14	2.82	-8.84	14.10	18.53
TFP_{post}^{growth}	-6.73	-5.22	-17.89	4.02	17.13
Debt maturing 2008	30.81	24.98	15.80	39.70	21.51
Observations	134,838				
Debt maturing 2008 (Non-GIIPS)	26.05	21.56	14.23	32.58	17.86
Debt maturing 2008 (GIIPS)	37.45	31.89	19.50	49.99	24.26
ΔCDS_c (Non-GIIPS)	90.59	86.48	71.05	92.25	40.00
ΔCDS_c (GIIPS)	94.11	88.68	87.41	105.70	10.05

ΔTFP_{growth} is the difference in the TFP growth rate pre- versus post-crisis. TFP_{pre}^{growth} is the average TFP growth rate pre-crisis. TFP_{post}^{growth} is the average TFP growth rate post-crisis. The precrisis period ranges from 2002 to 2007. The post-crisis period ranges from 2008 to 2013. The table also reports the amount of debt maturing in 2008 divided by average total sales precrisis for GIIPS and Non-GIIPS countries separately. ΔCDS_c is the change in the country-level CDS between the weeks before and after the Lehman bankruptcy, where the change in the country-level CDS is calculated as an average of the changes in domestic banks' CDS spread over the same window for GIIPS and non-GIIPS countries.

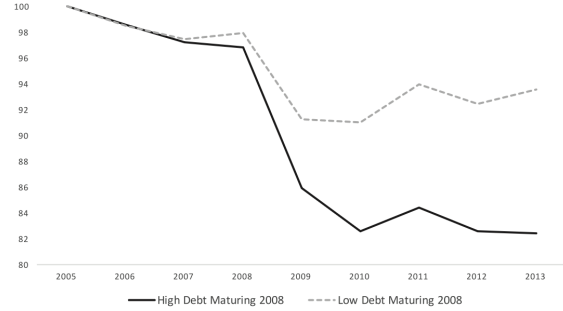


Figure 1
TFP level path for firms with different rollover risks (index 100 = 2005)
The TFP level path is shown as an index taking value of 100 in 2005. *High debt maturing 2008* corresponds to the 75th percentile of the distribution of *Debt maturing 2008*. *Low debt maturing 2008* corresponds to the 25th percentile of the distribution of *Debt maturing 2008*. *Debt maturing 2008* is the amount of debt maturing in 2008 divided by average total sales precrisis.

Read:

- Table 1 establishes the basic puzzle: the average firm's annual TFP growth falls from 2.14% before the crisis to -6.73% after it.
- The key descriptive treatment variable is economically large: the median firm has debt maturing in 2008 equal to roughly 25% of pre-crisis sales.
- Figure 1 is the visual pre-trend test. High- and low-rollover-risk firms move similarly before 2008, then diverge sharply after the crisis.
- The persistence matters most: the TFP level gap opens in 2009 and keeps widening through 2013, which fits the paper's "hysteresis" interpretation rather than a short cyclical dip.

5.2 Table 2 and Figure 2: Baseline result

Read:

- Table 2 is the core result. In the preferred specification with country-sector fixed effects and controls, the coefficient on debt maturing in 2008 is -0.0935.
- The paper interprets this as follows: a 10-percentage-point increase in maturing debt is associated with about a 0.94-percentage-point larger fall in annual post-crisis TFP growth.
- Figure 2 translates the coefficient into an intuitive bar chart comparing firms at the 25th and 75th percentiles of rollover risk. Both groups look similar pre-crisis, but vulnerable firms experience the much larger post-crisis collapse.
- The paper's back-of-the-envelope aggregation based on this table suggests that financial frictions may explain about 37% of the within-firm productivity slowdown in the sample.

Table 2
Baseline regression results

Dependent variable	(1)	(2)	(3)	(4)
	ΔTFP_{growth}			
Debt maturing 2008	−0.0693*** (0.007)	−0.0704*** (0.006)	−0.0674*** (0.006)	−0.0935*** (0.008)
R-squared	.127	.131	.142	.151
N	134,838	134,838	134,838	134,838
Country*Sector FEs	No	No	Yes	Yes
Sector FEs	No	Yes	—	—
Country FEs	Yes	Yes	—	—
Controls	No	No	No	Yes

The dependent variable ΔTFP_{growth} is the difference in the average TFP growth rate between pre- and post-crisis periods. *Debt maturing 2008* is the amount of debt maturing in 2008 divided by average total sales precrisis. The post-crisis period starts in 2008. Firm-specific controls include firm age, size of assets, and earnings (EBITDA). Standard errors are in parentheses. Standard errors are clustered at the country-sector level. * $p < .1$; ** $p < .05$; *** $p < .01$.

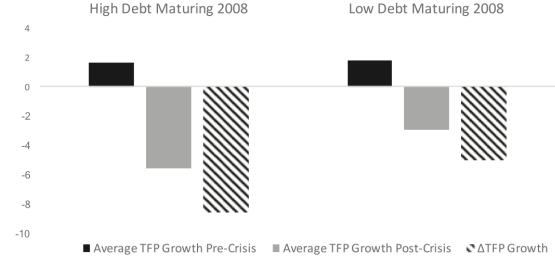


Figure 2
Estimated TFP growth decline for firms with different rollover risks
 ΔTFP_{growth} is the difference in the average TFP growth rate between the pre- and post-crisis periods. *Average TFP growth pre- (post) crisis* is the average TFP growth rate precrisis (post-crisis). *Debt maturing 2008* is the amount of debt maturing in 2008 divided by average total sales precrisis. *High (low) debt maturing 2008* corresponds to the 75th percentile (25th percentile) of the cross-firm distribution of *Debt maturing 2008*. The post-crisis sample starts in 2008. Column 4 of Table 2 provides the underlying regression estimates.

Table 3
Baseline regression: Controlling for short-term debt before the crisis

Dependent variable	(1)	(2)	(3)	(4)
	ΔTFP_{growth}			
Debt maturing 2008	−0.0686*** (0.007)	−0.0729*** (0.011)	−0.0953*** (0.008)	−0.0883*** (0.012)
Avg. current liabilities precrisis		0.00637 (0.011)		−0.0104 (0.011)
R-squared	.142	.142	.151	.151
N	130,319	130,319	130,319	130,319
Country*Sector FEs	Yes	Yes	Yes	Yes
Sector FEs	—	—	—	—
Country FEs	—	—	—	—
Controls	No	No	Yes	Yes

The dependent variable ΔTFP_{growth} is the difference in the average TFP growth rate between pre- and post-crisis periods. *Debt maturing 2008* is the amount of debt maturing in 2008 divided by average total sales precrisis. The post-crisis period starts in 2008. *Avg. current liabilities precrisis* is the mean of current liabilities divided by sales before 2007. Firm-specific controls include firm age, size of assets, and earnings (EBITDA). Standard errors are in parentheses. Standard errors are clustered at the country-sector level. * $p < .1$; ** $p < .05$; *** $p < .01$.

5.3 Table 3: Controlling for ordinary short-term debt

Read:

- Table 3 asks whether the main result is just picking up firms that generally use more short-term debt.
- Adding average pre-crisis current liabilities barely moves the main coefficient, and the extra current-liabilities control is statistically insignificant.
- The takeaway is that the problem is not “short-term debt in general” but debt maturing specifically in the Lehman year, exactly when rollover conditions collapsed.

5.4 Table 4 and Figure 3: Cross-country heterogeneity in the credit shock

Read:

- Table 4 interacts rollover risk with the country-level bank CDS jump around Lehman. The preferred interaction coefficient is -0.113 , strongly significant.
- This means vulnerable firms do worse precisely where credit conditions tighten more. That is the paper’s cleanest evidence that the mechanism is financial rather than merely cyclical.
- Figure 3 shows the same logic graphically: the gap in post-crisis TFP decline between high- and low-rollover-risk firms is much larger in countries with above-average CDS spikes.
- This is the exhibit that most clearly turns the baseline DD into a causal credit-supply story.

Table 4
Extended specification: Accounting for cross-country heterogeneity in exposure to the collapse of Lehman Brothers

Dependent variable	(1)	(2)	(3)	(4)
	ΔTFP_{growth}			
Debt maturing 2008	−0.0706*** (0.007)	−0.0956*** (0.007)	−0.0682*** (0.006)	−0.0951*** (0.007)
Debt maturing 2008 * ΔCDS_c	−0.0823*** (0.024)	−0.105*** (0.024)	−0.0824*** (0.020)	−0.113*** (0.021)
R-squared	.143	.161	.156	.169
N	104,275	104,275	104,275	104,275
Country*Sector FEs	No	No	Yes	Yes
Sector FEs	No	Yes	—	—
Country FEs	Yes	Yes	—	—
Controls	No	No	No	Yes

The dependent variable ΔTFP_{growth} is the difference in the average TFP growth rate between the pre- and post-crisis periods. *Debt maturing 2008* is the amount of debt maturing in 2008 divided by average total sales precrisis. The post-crisis period starts in 2008. ΔCDS_c is the standardized change in the country-level CDS between the weeks before and after the Lehman bankruptcy, where the change in the country-level CDS is calculated as an average of the changes in domestic banks' CDS spread over the same window. Firm-specific controls include firm age, size of assets, employment, and earnings (EBITDA). Each specification also includes interactions between each of these controls and ΔCDS_c . Standard errors are in parentheses. Standard errors are clustered at the country-sector level. * $p < .1$; ** $p < .05$; *** $p < .01$.

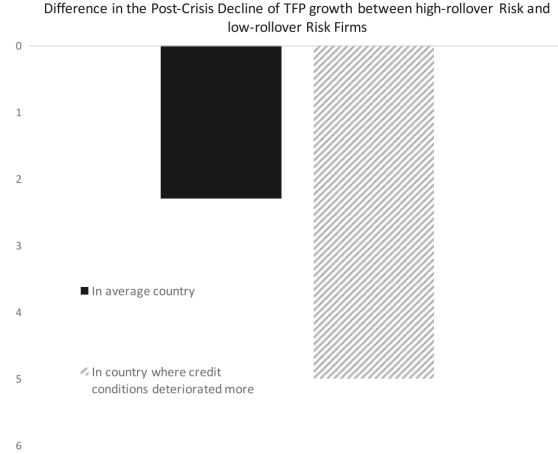


Figure 3
Estimated TFP growth decline for firms with different rollover risks: The role of country exposure in the collapse of Lehman Brothers
Rollover risk is the amount of debt maturing in 2008 divided by average total sales precrisis. *High (low) debt maturing 2008* corresponds to the 75th (25th) percentile of the cross-firm distribution of *Debt maturing 2008*. The *average country* corresponds to a no change in CDS spread after standardizing the variable. The *country where credit conditions deteriorated more* corresponds to 1-standard-deviation larger change in standardized CDS spread compared to the average country CDS spreads. The post-crisis sample starts in 2008. Column 4 of Table 4 provides the underlying regression estimates.

5.5 Table 5: Firm-bank matched exposure

Read:

- Table 5 repeats the exercise at the firm-bank level using the CDS performance of a firm's own creditor banks.
- The preferred interaction coefficient is about -0.0229 , again negative and significant, despite the much smaller sample.
- This is persuasive because it uses within-country variation in creditor stress. The result says that not only vulnerable firms but specifically vulnerable firms tied to harder-hit banks suffer the largest productivity losses.
- The economically relevant caveat is the sample reduction: only 20,798 firms remain once bank matching and bank CDS data are required.

Table 5
Extended specification: Accounting for firm-level heterogeneity in exposure to the collapse of Lehman Brothers

Dependent variable	(1)	(2)	(3)	(4)
	ΔTFP_{growth}			
Debt maturing 2008	−0.112*** (0.014)	−0.154*** (0.014)	−0.114*** (0.015)	−0.163*** (0.015)
ΔCDS_i	−0.140 (0.214)	−0.251 (0.214)	−0.176 (0.217)	−0.301 (0.214)
Debt maturing 2008 * ΔCDS_i	−0.0232** (0.010)	−0.0231** (0.010)	−0.0243** (0.010)	−0.0229** (0.011)
R-squared	.0640	.0922	.0793	.109
N	20,798	20,798	20,798	20,798
Country*Sector FEs	No	No	No	Yes
Sector FEs	No	No	Yes	—
Country FEs	Yes	Yes	Yes	—
Controls	Yes	Yes	Yes	Yes

The dependent variable ΔTFP_{growth} is the difference in the average TFP growth rate between pre- and post-crisis periods. *Debt maturing 2008* is the amount of debt maturing in 2008 divided by average total sales precrisis. The post-crisis period starts in 2008. ΔCDS_i refers to the standardized change in the average CDS spread of the firm's main creditor bank(s) (up to five of them, drawn from the *BANKER* variable in AMADEUS) between the weeks before and after the collapse of Lehman Brothers. Firm-specific controls include firm age, size of assets and earnings (EBITDA) and are interacted with ΔCDS_i . Standard errors are in parentheses. Standard errors are clustered at the country-sector level. * $p < .1$; ** $p < .05$; *** $p < .01$.

Table 6
Financial frictions and innovation

Dependent variable	(1)	(2)	(3)	(4)	(5)	(6)
	$\Delta Int_Investment$		$\Delta Share_Intangible$		$\Delta PatentApplications$	
Debt maturing 2008	−0.0188*** (0.002)	−0.0184*** (0.002)	−0.0633*** (0.010)	−0.0584*** (0.010)	−0.00137*** (0.00037)	−0.00100*** (0.00035)
R-squared	.0406	.0407	.373	.379	.0353	.0419
N	97,487	97,487	101,150	101,150	37,136	37,136
Country*Sector FEs	Yes	Yes	Yes	Yes	Yes	Yes
Controls	No	Yes	No	Yes	No	Yes

The dependent variable $\Delta Int_Investment$ for Columns 1 and 2 is the difference in the investment in intangible assets as a ratio of the value added pre- versus post-crisis. The dependent variable $\Delta Share_Intangible$ for Columns 3 and 4 is the difference in the share of intangible assets in total capital pre- versus post-crisis. The dependent variable $\Delta PatentApplications$ for Columns 5 and 6 is the difference in the average number of patent applications each year pre- versus post-crisis. *Debt maturing 2008* is the amount of debt maturing in 2008 divided by average total sales precrisis. The post-crisis period starts in 2008. Firm-specific controls include firm age, size of assets, and earnings (EBITDA). Standard errors are in parentheses. Standard errors are clustered at the country-sector level. * $p < .1$; ** $p < .05$; *** $p < .01$.

5.6 Table 6: Innovation channel

Read:

- Table 6 is the mechanism table. Vulnerable firms cut intangible investment more after the crisis, with coefficients around -0.018 on intangible investment rates and around -0.058 on the share of intangible assets.
- The patent results are also negative and significant, indicating weaker innovation outcomes, not just lower accounting investment in intangibles.
- This is where the paper earns the “TFP hysteresis” label: temporary credit stress appears to lower the path of intangible accumulation and thus the path of future productivity.

5.7 Table 7 and Figure 4: Placebo around the 2000 recession

Table 7
Placebo test: Early 2000s recession

Dependent variable	(1) ΔTFP_{growth} (post-2000 minus pre-2000)	(2) ΔTFP_{growth} (post-2000 minus pre-2000)	(3) $\Delta Int_Investment$ (post-2000 minus pre-2000)	(4) $\Delta Int_Investment$ (post-2000 minus pre-2000)
Debt maturing 2000	-0.0719 (0.046)	-0.0152 (0.031)	0.00483 (0.033)	0.00496 (0.028)
R-squared	.170	.204	.104	.105
N	53,139	53,139	3,295	3,295
Country*Sector FEs	Yes	Yes	Yes	Yes
Controls	No	Yes	No	Yes

The placebo post-crisis period runs from 2000 to 2005, with 2000 assumed to be the crisis year. The dependent variable ΔTFP_{growth} is the difference in the average TFP growth rate between the pre- and post-crisis periods. The dependent variable $\Delta Int_Investment$ is the difference in the investment in intangible assets as a ratio of value added pre- versus post-crisis. *Debt maturing 2000* is the amount of debt maturing in 2000 divided by average total sales pre-2000. Firm-specific controls include firm age, size of assets, and earnings (EBITDA). Standard errors are in parentheses. Standard errors are clustered at the country-sector level. * $p < .1$; ** $p < .05$; *** $p < .01$.

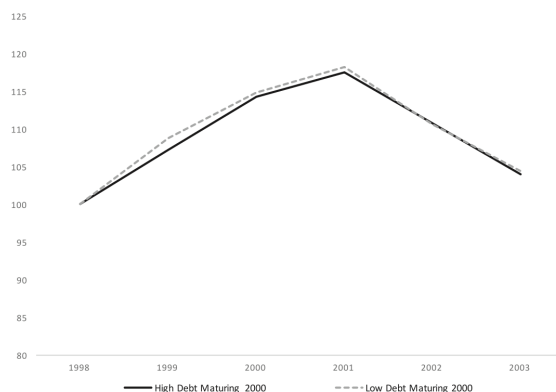


Figure 4
TFP level path for firms with different rollover risks: 2000 recession (index 100 = 1998)
The TFP level path is shown as an index taking value 100 in 1998. *High (low) debt maturing in 2000* corresponds to the 75th (25th) percentile of the cross-firm distribution of *High debt maturity 2000*. *Debt maturing 2000* is the amount of debt maturing in 2000 divided by average total sales pre-2000.

Read:

- Table 7 is the main falsification test. When the authors pretend the crisis year is 2000, debt maturing in 2000 does not predict subsequent TFP or intangible-investment changes.
- Figure 4 mirrors Figure 1 but for the early-2000s recession and shows no post-2000 divergence between high- and low-rollover-risk firms.
- This is powerful because it shows the baseline result is not a generic feature of recessions or of debt maturity structure. It is specific to a recession with a genuine financial freeze.

5.8 Tables 8 and 9: Measurement and rollover-risk robustness

Read:

- Table 8 shows the result survives alternative productivity measures: labor productivity, a Cobb-Douglas constant-returns Solow residual, and OLS-based TFP all deliver the same negative pattern.
- Table 9 addresses the short-term-debt objection in three different ways: exclude trade credit, keep only debt with maturity of at least two years, and drop firms with only short-term debt.

Table 8
Baseline regression: Alternative measures of productivity

Dependent variable	(1) Labor Productivity	(2) Cobb-Douglas CRS	(3) Cobb-Douglas CRS	(4) Cobb-Douglas CRS	(5) OLS	(6) OLS
Debt maturing 2008	−0.0438*** (0.004)	−0.0513*** (0.004)	−0.0703*** (0.008)	−0.106*** (0.010)	−0.0611*** (0.008)	−0.0950*** (0.010)
R-squared	.0297	.0342	.0864	.0975	.0927	.102
N	143,472	143,472	145,400	145,400	142,734	142,734
Country*Sector FEs	Yes	Yes	Yes	Yes	Yes	Yes
Controls	No	Yes	No	Yes	No	Yes

The dependent variable *Labor productivity* is the difference in the labor productivity growth rate pre- versus post-crisis. *Cobb-Douglas CRS* is the difference in TFP (Solow residual) pre- versus post-crisis, assuming industry-specific labor shares from the OECD-STAN data and Cobb-Douglas production function with constant returns to scale (see Gal 2013 for further details). *OLS* is the difference in TFP derived from an OLS regression, based on a Cobb-Douglas production function without additional constraints on returns to scale. *Debt maturing 2008* is the amount of debt maturing in 2008 divided by average total sales precrisis. Post-crisis starts in 200. Firm-specific controls include firm age, size of assets and earnings (EBITDA). Standard errors are in parentheses. Standard errors are clustered at the country-sector level. * $p < .1$; ** $p < .05$; *** $p < .01$.

Table 9
Baseline regression: Alternative measures of debt maturing 2008

Dependent variable	(1)	(2)	(3) ΔTFP^{growth}	(4)
Debt maturing 2008	−0.0935*** (0.008)		−0.0988*** (0.013)	−0.0906*** (0.009)
Debt maturity 2008, excl. trade credit		−0.0822*** (0.007)		
R-squared	.151	.150	.135	.162
N	134,838	134,295	31,285	103,409
Country*Sector FEs	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes
Sample	All	All	Debt maturity >=2 years	Excl. firms with only ST debt

The dependent variable ΔTFP^{growth} is the difference in the average TFP growth rate between pre- and post-crisis periods. *Debt maturing 2008* is the amount of debt maturing in 2008 divided by average total sales precrisis. *Debt maturity 2008 excl. trade credit* removes debt to suppliers and contractors (trade credit) from the total debt that matures in 2008. Column 3 only includes firms that did not issue any financial debt in 2007. Column 4 excludes all firms that we know only have short-term debt. The post-crisis period starts in 2008. Firm-specific controls include firm age, size of assets and earnings (EBITDA). Standard errors are in parentheses. Standard errors are clustered at the country-sector level. * $p < .1$; ** $p < .05$; *** $p < .01$.

- Across all these variants, the coefficient on debt maturing in 2008 stays close to the baseline. That gives the rollover-risk interpretation a lot of credibility.

5.9 Tables 10 and 11: Other vulnerabilities and cross-country heterogeneity

Table 10
Baseline regression: Incorporating other measures of financial vulnerability

Dependent variable	(1)	(2) ΔTFP^{growth}	(3)	(4)
Debt maturing 2008	−0.0900*** (0.007)	−0.0907*** (0.007)	−0.0917*** (0.007)	−0.0907*** (0.007)
Cash precrisis	0.0284*** (0.007)		0.000564 (0.008)	0.000564 (0.008)
Leverage precrisis		−0.0363*** (0.008)		−0.0229*** (0.009)
ICR precrisis			−0.0236*** (0.005)	−0.0193*** (0.005)
R-squared	.151	.151	.158	.158
N	133,272	134,838	117,882	116,441
Country*Sector FEs	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes

The dependent variable ΔTFP^{growth} is the difference in the average TFP growth rate between the pre- and post-crisis periods. *Cash precrisis* is the ratio of average cash and cash equivalents to total assets before the crisis. *Leverage precrisis* is average leverage, measured as the debt-to-asset ratio, before the crisis. *ICR precrisis* is the average ratio of interest expenses to earnings (EBITDA), that is, the inverse of the interest coverage ratio, before the crisis. Firm-specific controls include firm age, size of assets and earnings (EBITDA). Standard errors are in parentheses. Standard errors are clustered at the country-sector level. * $p < .1$; ** $p < .05$; *** $p < .01$.

Table 11
Baseline regression: Country groups

	(1) GIIPS	(2) Non-GIIPS	(3) Euro	(4) Non-Euro
Debt maturing 2008	−0.132*** (0.014)	−0.0517*** (0.009)	−0.116*** (0.011)	−0.0492*** (0.012)
R-squared	.0957	.0737	.103	.0709
N	83,714	61,686	112,003	33,397
Country*Sector FEs	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes

The dependent variable ΔTFP^{growth} is the difference in the average TFP growth rate between the pre- and post-crisis periods. *Debt maturing 2008* is the amount of debt maturing in 2008 divided by average total sales precrisis. Post-crisis starts in 2008. Firm-specific controls include firm age, size of assets and earnings (EBITDA). Column 1 considers firms in GIIPS countries only (Greece, Ireland, Italy, Portugal, and Spain), and Column 2 excludes firms in GIIPS countries. Column 3 considers European firms only, whereas Column 4 considers non-European firms only (Japan, Korea, the United Kingdom, and Sweden). Standard errors are in parentheses. Standard errors are clustered at the country-sector level. * $p < .1$; ** $p < .05$; *** $p < .01$.

Read:

- Table 10 shows that cash, leverage, and interest-coverage ratios also matter in expected directions, but they do not wash out the effect of debt maturing in 2008.
- Table 11 shows much larger effects in peripheral euro-area countries: the coefficient is −0.132 in the GIIPS/periphery group versus −0.0517 in the non-GIIPS group.
- The authors' aggregation implies that financial frictions may explain roughly 46% of the within-firm TFP slowdown in the GIIPS/periphery group but only about 15% in the rest.
- This matters for external interpretation: the paper is not saying finance mattered equally everywhere. It mattered most where the banking shock was most intense.

5.10 Tables 12 and 13: Firm-age split and conditioning on initial productivity

Read:

- Table 12 shows similar negative effects for older and younger firms, so the result is not driven only by young firms being forced into short debt.

Table 12
Baseline regression: Old firms versus young firms

Dependent variable	(1)	(2)	(3)	(4)
	$\Delta TFPgrowth$			
	Old firms	Young firms	Old firms	Young firms
Debt maturing 2008	-0.0647*** (0.009)	-0.0725*** (0.008)	-0.0823*** (0.01)	-0.105*** (0.009)
R-squared	.183	.127	.193	.138
N	48,827	85,859	48,827	85,859
Country*Sector FEs	Yes	Yes	Yes	Yes
Controls	No	No	Yes	Yes

The dependent variable $\Delta TFPgrowth$ is the difference in the average TFP growth rate between the pre- and post-crisis periods. Old (young) firms are firms with the age older (less) than 16 years in 2007. Firm-specific controls include firm age, size of assets and earnings (EBITDA). Standard errors are in parentheses. Standard errors are clustered at the country-sector level. * $p < .1$; ** $p < .05$; *** $p < .01$.

Table 13
Controlling for the precrisis TFP level

Dependent variable	(1)	(2)	(3)	(4)
	$\Delta TFPgrowth$			
Debt maturing 2008	-0.0698*** (0.006)	-0.0670*** (0.005)	-0.0629*** (0.005)	-0.0899*** (0.006)
Average TFP level precrisis	-0.0641 (0.419)	0.637 (0.655)	0.860 (0.673)	0.595 (0.609)
R-squared	.127	.131	.142	.151
N	134,838	134,838	134,838	134,838
Country*Sector FEs	No	No	Yes	Yes
Sector FEs	No	Yes	—	—
Country FEs	Yes	Yes	—	—
Controls	No	No	No	Yes

The dependent variable $\Delta TFPgrowth$ is the difference in the average TFP growth rate between the pre- and post-crisis periods. Average TFP level precrisis is the average firm-level TFP level (measured by the Wooldridge method) before the crisis. Firm-specific controls include firm age, size of assets, employment, and earnings (EBITDA). * $p < .1$; ** $p < .05$; *** $p < .01$.

- Table 13 adds the pre-crisis TFP level and finds it is insignificant, while the main debt-maturity coefficient is basically unchanged.
- Together these tables reassure the reader that the result is not just mean reversion from initially low-productivity firms or a peculiarity of very young firms.

6 Short summary

- **Conclusion.** Firms with more debt coming due right when Lehman hit suffered a much larger and longer-lasting post-crisis productivity slowdown.
- **Intuition.** When rollover became difficult, fragile firms cut intangible and innovative activity, and that temporary financial distress pushed down the future productivity path.
- **Empirical lesson.** The combination of predetermined debt maturity, narrow-window bank CDS shocks, within-country-sector comparisons, and placebo tests makes a compelling case that credit supply can damage productivity inside firms, not just employment or investment.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- **Main finding.** Financial frictions contributed materially to the post-2008 productivity slowdown by depressing within-firm TFP growth.
- **Magnitude.** In the baseline, 10 percentage points more debt maturing in 2008 predict roughly 0.94 percentage points lower annual post-crisis TFP growth; the aggregate calculation suggests finance explains about one-third of the within-firm slowdown.
- **Credit-supply evidence.** The effect is stronger both in countries with larger bank-CDS spikes and for firms borrowing from more stressed banks.
- **Mechanism evidence.** Vulnerable firms cut intangible investment and patenting more, consistent with the idea that crises hit innovation especially hard because it is difficult to collateralize.
- **Persistence.** The TFP gap widens for years after the initial shock, so the paper argues for hysteresis rather than a transitory downturn.

- **Cross-country implication.** The effect is especially large in crisis-hit peripheral euro-area economies, suggesting that banking stress can translate into persistent productivity weakness at the macro level.

7.2 Economic intuition

- A firm with a lot of debt maturing in 2008 needed refinancing exactly when banks became suddenly unwilling or unable to lend.
- Faced with a financing squeeze, the firm cuts expenditures with low pledgeability and slow payoff, especially intangible investment.
- Because intangible capital affects future efficiency and innovation, the damage shows up not only in contemporaneous activity but also in lower future TFP growth.
- The mechanism therefore links a short-lived credit market freeze to a long-lived productivity slowdown.

7.3 Takeaway

This is a strong paper because it takes a macro puzzle—the post-crisis productivity slowdown—and explains part of it with a clean firm-level credit design. The contribution is not just that leverage matters in crises. It is that the maturity structure of liabilities interacts with a sudden credit freeze to create persistent within-firm productivity losses, plausibly by choking off intangible investment and innovation. For reading purposes, the key things to remember are the debt-maturing-in-2008 design, the Lehman-window bank-CDS interactions, and the idea of *TFP hysteresis*.